

A MINIMAL-DEGREE (23, 4)-MODEL AND AN ARITHMETIC PROGRESSION OF MATHIEU M_{23} -POLYNOMIALS

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ABSTRACT. Huang, Jackson, Lee, Poonen, Pries, and Zhang recently constructed a regular M_{23} -extension of $\mathbf{Q}(t)$ by means of an explicit polynomial of bidegree (23, 8). They observed that the same genus-4 curve admits a rational function of degree 4, and that no rational function of degree at most 3 exists over $\mathbf{Q}$, but did not compute the resulting equation. We make their observation explicit. Starting from their published equation, we compute two canonical differentials whose ratio is a degree-4 rational function and obtain a primitive polynomial

$$P(T, W) \in \mathbf{Z}[T, W], \quad \deg_W P = 23, \quad \deg_T P = 4.$$

An exact identity in the original function field and irreducibility modulo 31 prove that P defines precisely the same regular M_{23} -cover. We provide the equation, the rational function effecting the change of model, and exact Singular, SageMath, and Magma certificates. Moreover, for every integer $t \equiv 2 \pmod{319}$, the primitive part of $P(t, x)$ has Galois group exactly M_{23} over $\mathbf{Q}$. Irreducibility modulo 29 and factor degrees (2, 7, 14) modulo 11 give a uniform two-prime certificate for the entire arithmetic progression. Controlled specialization near the rational branch point then produces infinitely many distinct, mutually independent M_{23} -extensions within this progression.

1. INTRODUCTION

Let $F(T, V) \in \mathbf{Q}(T)[V]$ be the degree-23 polynomial constructed in [1]. Its splitting field over $\mathbf{Q}(T)$ is a regular extension with Galois group M_{23} . If $D = T^2 + 23$, then the cleared integral model $\Phi = D^4 F \in \mathbf{Z}[T, V]$ has bidegree (23, 8). Their Remark 3.4 proves that the source curve possesses a degree-4 function defined over $\mathbf{Q}$, and that no function of degree at most 3 is defined over $\mathbf{Q}$. The degree-4 function and its equation were left implicit.

This note computes that degree-4 function and proves a uniform specialization theorem. Here “minimal degree” refers to the degree of a nonconstant $\mathbf{Q}$ -rational map from the source curve to $\mathbf{P}^1$. Our two main results are the following.

Theorem 1.1. *Let $X \rightarrow \mathbf{P}_{\mathbf{Q}}^1$ be the degree-23 cover defined by the polynomial $F(T, V)$ of [1]. There are explicit polynomials $J_0, J_1 \in \mathbf{Z}[T, V]$ and a primitive irreducible polynomial $P \in \mathbf{Z}[T, W]$ such that*

$$\deg_W P = 23, \quad \deg_T P = 4,$$

and, in $\mathbf{Q}(X)$,

$$W = \frac{J_0(T, V)}{J_1(T, V)}, \quad P(T, W) = 0.$$

Moreover,

$$\mathbf{Q}(T, W) = \mathbf{Q}(T, V).$$

Consequently the splitting field of $P(T, W)$ over $\mathbf{Q}(T)$ is a regular M_{23} -extension. The projection $X \rightarrow \mathbf{P}_W^1$ has degree 4, which is minimal among nonconstant rational maps $X \rightarrow \mathbf{P}^1$ defined over $\mathbf{Q}$.

Date: Draft of August 20, 2026.

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The arithmetic counterpart is a single congruence class all of whose specializations have the full Mathieu group.

Theorem 1.2. *For every integer*

$$t \equiv 2 \pmod{319},$$

the primitive part of $P(t, x)$ has degree 23, and its splitting field over $\mathbf{Q}$ has Galois group M_{23} in its natural degree-23 action.

To the best of our knowledge, as of the August 2026 search described in the supplement and relative to arXiv:2608.08538v1, this is the first explicit computation of the minimal degree-4 function anticipated in [1, Remark 3.4], and the first displayed arithmetic progression of specializations of that cover for which every member is uniformly certified to have Galois group M_{23} . This priority statement concerns the explicit equation and congruence data, not the underlying specialization method.

The full 120-term polynomial is supplied electronically in three equivalent formats. It has largest coefficient of 71 decimal digits. Printing it in the body would obscure the construction; a checksum and selected coefficients appear in section 5.

The geometric argument has two steps. The adjoint calculation in section 2 finds the degree-4 pencil directly on the singular plane model. The short exact certificate in section 4 then proves the result without requiring the reader to trust the modular discovery process. More precisely, the adjoint and CRT calculations are the discovery path; theorem 1.1 depends only on the exact function-field identity and irreducibility below, together with the monodromy and minimality results of [1]. Section 6 proves theorem 1.2 by a uniform two-prime Frobenius certificate, extracts one completely explicit polynomial by taking $t = 2$, and uses controlled ramification to obtain infinitely many mutually independent M_{23} -extensions of $\mathbf{Q}$.

The last infinitude statement is an explicit extraction result, not a new existence theorem. Abstractly, infinitely many mutually independent M_{23} -extensions already follow from the regular cover [1, Corollary 1.4(d)]; the refinement here keeps every field inside one displayed congruence class and supplies a new ramified prime on demand.

2. THE PLANE MODEL AND ITS ADJOINTS

Retain $D = T^2 + 23$ and $\Phi(T, V) = D^4 F(T, V) \in \mathbf{Z}[T, V]$. Then $\deg_V \Phi = 23$ and $\deg_T \Phi = 8$. We write C_0 for the affine plane curve $\Phi = 0$ and X for its normalization. The compactification under consideration is the closure in $\mathbf{P}_T^1 \times \mathbf{P}_V^1$.

2.1. The two points above $T^2 + 23 = 0$. Let $b, c \in X(\overline{\mathbf{Q}})$ be the unique points over the roots of D . They are conjugate over $\mathbf{Q}$ and each has ramification index 23 for the T -map [1, Section 3].

Lemma 2.1 (Valuations at the two ramification points). *At either $q \in \{b, c\}$ one has*

$$\text{ord}_q(T - T(q)) = 23, \quad \text{ord}_q(D) = 23, \quad \text{ord}_q(V) = -4,$$

and

$$\text{ord}_q\left(\frac{dT}{\Phi_V}\right) = 18.$$

Proof. The first two equalities follow from total ramification and the fact that D has a simple zero at $T(q)$. The coefficients of Φ , viewed as a polynomial in V , have a Newton-polygon edge joining $(0, 0)$ and $(23, 4)$ at $D = 0$. Hence its unique branch has $\text{ord}_q(V) = -4$.

For completeness, the terms of Φ_V on this branch have orders

$$23 \text{ord}_D(ia_i) - 4(i - 1),$$

where $a_i(T)$ is the coefficient of V^i . The minimum is 4, attained uniquely by differentiating the D^4V^{23} term; all other orders are at least 5. Thus $\text{ord}_q(\Phi_V) = 4$. Finally $\text{ord}_q(dT) = 23 - 1 = 22$, so $\text{ord}_q(dT/\Phi_V) = 22 - 4 = 18$. $\square$

The calculation is also reproduced directly from the normalized coefficient table by

`verification/verify_boundary_valuations.gp.`

For a polynomial $A(T, V)$ consider the adjoint differential

$$\omega_A = \frac{A(T, V) dT}{\Phi_V(T, V)}.$$

The interior of the bidegree-(8, 23) rectangle gives the usual support $0 \leq i \leq 6$, $0 \leq j \leq 21$ for monomials T^iV^j . Regularity at b and c sharpens this $7 \cdot 22 = 154$ -dimensional space substantially.

Lemma 2.2 (Adjoints regular at b and c). *The differentials ω_A having interior support and regular at both b and c are represented by the 88-dimensional space*

$$\begin{aligned} \mathcal{A} = & \bigoplus_{j=0}^4 V^j \mathbf{Q}[T]_{\leq 6} \oplus \bigoplus_{j=5}^{10} DV^j \mathbf{Q}[T]_{\leq 4} \\ & \oplus \bigoplus_{j=11}^{16} D^2V^j \mathbf{Q}[T]_{\leq 2} \oplus \bigoplus_{j=17}^{21} D^3V^j \mathbf{Q}. \end{aligned}$$

In particular,

$$\dim \mathcal{A} = 35 + 30 + 18 + 5 = 88.$$

Proof. Write $A = \sum_{j=0}^{21} a_j(T)V^j$. By Lemma 2.1, a term $D^m c(T)V^j dT/\Phi_V$ has order

$$23m - 4j + 18$$

at b and c , unless c also vanishes there. Regularity therefore requires

$$m \geq \left\lceil \frac{4j - 18}{23} \right\rceil,$$

giving respectively $m = 0, 1, 2, 3$ on the four displayed blocks. Terms having distinct exponents j cannot cancel: their orders are distinct modulo 23. Divisibility by D^m lowers the residual T -degree from 6 to $6 - 2m$. Counting coefficients gives the stated dimension. $\square$

2.2. The affine nodes and the canonical system. At an ordinary node, Rosenlicht's adjoint condition [2] is the vanishing of the numerator A . Thus each node contributes one linear evaluation condition. Independence of the resulting conditions is not inferred from the node count; it is the rank computation recorded below. We impose the conditions in one stroke using the reduced singular scheme modulo 31.

Exact computation 2.3 (The node graph modulo 31). Over $\mathbf{F}_{31}$ the Jacobian ideal

$$(\Phi, \Phi_T, \Phi_V)$$

has lexicographic basis

$$h_{84}(T), \quad V - s(T),$$

where $\deg h_{84} = 84$ and h_{84} is squarefree. The determinant of the Hessian is a unit on this scheme. Thus the affine singular scheme consists geometrically of 84 reduced ordinary nodes.

This statement is certified by `verification/verify_nodes_mod31.sing`; the explicit h_{84} and s are in `data/node_graph_mod31.inc`.

Evaluation of the basis in Lemma 2.2 on the node graph gives an 84×88 matrix over $\mathbf{F}_{31}$.

Exact computation 2.4 (Canonical kernel and degree-4 pencil). The node-evaluation matrix has rank 84, so its kernel has dimension 4. This is the canonical adjoint space $H^0(X, K_X)$. On this kernel, vanishing at $b + c$ has rank 2, leaving the two-dimensional space $H^0(X, K_X - b - c)$ and hence a pencil.

Explanation of the two conditions at b and c . Among the blocks of Lemma 2.2, the only term that can have order zero at b or c is

$$D^2(c_0 + c_1T + c_2T^2)V^{16}.$$

Its vanishing at both roots of D is equivalent to divisibility of the quadratic coefficient by D , namely

$$c_1 = 0, \quad c_0 = 23c_2.$$

These two conditions have rank 2 on the four-dimensional node kernel. The ranks in the computation are reproduced by `verification/verify_adjoint_mod31.py`. $\square$

The successive dimensions are therefore

$$154 \longrightarrow 88 \longrightarrow 4 \longrightarrow 2.$$

The last two arrows are certified by full-rank minors modulo 31. Since their ranks are already maximal possible, the same ranks hold in characteristic zero.

If ω_0 and ω_1 form a basis of $H^0(X, K_X - b - c)$, then their ratio $W = \omega_0/\omega_1$ is a rational function on X . The resulting row-reduced echelon-form (RREF) normalized pencil was reconstructed over $\mathbf{Q}$ by the Chinese remainder theorem. The least common multiple of the reconstructed coefficient denominators is 91; clearing it gives the polynomials $J_0, J_1 \in \mathbf{Z}[T, V]$ in the supplement. Their degrees in V are 20 and 21, respectively, and each has degree 6 in T .

3. RECONSTRUCTION OF THE BIDEGREE (23,4) EQUATION

Set $W = J_0/J_1$. Eliminating V between $\Phi(T, V)$ and $J_0(T, V) - WJ_1(T, V)$ means extracting the desired factor from

$$\text{Res}_V(\Phi(T, V), J_0(T, V) - WJ_1(T, V)).$$

We performed this calculation modulo good primes in a stable RREF normalization, reconstructed the coefficients, and then discarded the numerical evidence in favor of the exact certificate below.

For transparency, the equation used a 575-bit CRT modulus. All 120 coefficients reconstructed and reduced correctly at the independent holdout prime 500015017. The pencil used a 604-bit modulus and passed independent holdout prime 137. These holdouts are evidence about reconstruction, not ingredients in the proof of theorem 1.1; their complete records are `data/f4_crt_reconstruction.json` and `data/pencil_crt_reconstruction.json`. As an auxiliary geometric check, the reduction of P modulo 31 divides the displayed defining resultant; this is verified by `verification/verify_f4_mod31.sing`.

4. EXACT CERTIFICATE

Write

$$P(T, W) = \sum_{j=0}^{23} p_j(T)W^j.$$

The following exact computations form a compact certificate.

Proposition 4.1 (Exact function-field identity). *In $\mathbf{Q}(T)[V]/(\Phi)$ one has*

$$\sum_{j=0}^{23} p_j(T) J_0^j J_1^{23-j} = 0.$$

Equivalently, $P(T, J_0/J_1) = 0$ in $\mathbf{Q}(X)$.

Proof. This is a polynomial identity modulo Φ , equivalently modulo the monic degree-23 relation F over $\mathbf{Q}(T)$. The supplied Singular certificate evaluates the left side by homogeneous Horner reduction and obtains zero. The SageMath certificate independently rebuilds Φ, P, J_0, J_1 from JSON coefficient tables, reads no Singular inputs, and performs the same calculation in its own polynomial arithmetic. A self-contained Magma certificate reconstructs the same four inputs from their canonical coefficient tables and repeats the quotient-ring identity. All computations are exact. $\square$

Proposition 4.2 (Irreducibility). *The polynomial P is primitive and irreducible in $\mathbf{Q}[T, W]$. In particular it is irreducible in $\mathbf{Q}(T)[W]$.*

Proof. The coefficient content is 1. Its reduction modulo 31 retains bidegree $(23, 4)$ and is irreducible in $\mathbf{F}_{31}[T, W]$, as checked by Singular, SageMath, and Magma. The Magma computation supplies a separate native multivariate factorization backend. Gauss’s lemma proves the claim. $\square$

Proof of Theorem 1.1. By Proposition 4.1, $\mathbf{Q}(T, W)$ is a subfield of $\mathbf{Q}(T, V)$. By Proposition 4.2,

$$[\mathbf{Q}(T, W) : \mathbf{Q}(T)] = 23.$$

The published equation also gives $[\mathbf{Q}(T, V) : \mathbf{Q}(T)] = 23$. Hence the inclusion is an equality. The regular M_{23} monodromy follows from [1, Proposition 3.5 and Corollary 3.6].

Now regard P as a polynomial in T over $\mathbf{Q}(W)$. Since P is irreducible and has positive T -degree, it is primitive in $\mathbf{Q}[W][T]$. Gauss’s lemma and $\deg_T P = 4$ therefore give $[\mathbf{Q}(T, W) : \mathbf{Q}(W)] = 4$. Thus W has degree 4 on X . Minimality is the nonsplit-quadric argument of [1, Remark 3.4]: every degree at most 3 map on the geometric genus-4 curve comes from a ruling of its canonical quadric, whereas those rulings are not defined over the relevant quadratic field, and hence not over $\mathbf{Q}$. $\square$

5. EXACT DATA AND REPRODUCIBILITY

The paper, the $(23, 4)$ equation, and the PARI/GP, Singular, SageMath, and Magma certificates are available at

<https://github.com/research-reflexivity/frontier-math-m23-optimal-model>.

The cleared source model Φ is encoded canonically as the integer coefficient table `data/Fint_coefficients_Z.json`. It represents

$$\Phi(T, V) = (T^2 + 23)^4 F(T, V)$$

and is rendered deterministically into PARI/GP syntax. The mathematical input comes from the model of [1]; the exact upstream state is commit `f9c59eba2017090c5ea20f6f4e4a412055443463`. The supplement records the source hash and normalization without redistributing the upstream program.

The coefficient convention in `data/optimal_23_4_Z.json` is

$$\text{coefficients_T.then_W}[i][j] = [T^i W^j]P.$$

The coefficient of $T^4 W^{23}$ is

$$838610528294018704285119511337682389342017079161221371107109332287875.$$

The polynomial has 120 nonzero coefficients and coefficient content 1. It is supplied in JSON, PARI/GP, and Singular include formats. The exact pencil is represented analogously in `data/optimal_degree4_pencil.json` and in Singular include format. The primary SHA-256 fingerprints are

```
data/optimal_23_4_Z.json
cfb4185b2800744d524d87b3f08f5459cd38d42afb3787c11fa1e2aa3f25ee84
data/optimal_degree4_pencil.json
fdf571e4f47effc527e1726bdd99472be2c5e3390e45133fb3d7a3aab592619d
```

The complete scientific-data manifest is `CHECKSUMS.sha256`.

A clean verification consists of

```
make verify
make qa
```

or the convenience command `scripts/reproduce.sh`. The driver `verification/verify_optimal_23_4.py` checks the exact identity and finite-field certificate, while an independent SageMath path reconstructs every input from JSON and reads no Singular source. The optional `verification/verify_optimal_23_4.m` path repeats both core checks in Magma from a self-contained, deterministically generated input. Separate programs check the valuations at b and c , adjoint ranks, progression, and the ramification witness. Exact software versions are in `verification/environment.txt`; hashes, holdout records, and machine-readable summaries are included in the supplement.

6. A CERTIFIED ARITHMETIC PROGRESSION OVER $\mathbf{Q}$

We first record the specialization fact that identifies the ambient group.

Lemma 6.1 (Specialization subgroup). *Let $Q(T, x) \in \mathbf{Q}[T, x]$ have splitting field $L/\mathbf{Q}(T)$ with Galois group G . Suppose that $t \in \mathbf{Q}$ is a value at which the leading coefficient and the discriminant of Q with respect to x are nonzero. Then the splitting field of $Q(t, x)$ over $\mathbf{Q}$ has Galois group isomorphic, in its action on the specialized roots, to a subgroup of G .*

Proof. Work over the discrete valuation ring $R = \mathbf{Q}[T]_{(T-t)}$. By hypothesis the leading coefficient and discriminant are units of R . After dividing by the leading coefficient, Q is monic and still has unit discriminant. Hence the normalization in L is unramified above $T = t$. Choose a prime above $T = t$. Its residue extension has Galois group equal to its decomposition group, which is a subgroup of G . Reduction sends the generic roots to the distinct roots of $Q(t, x)$. These roots generate the residue extension: an element of the decomposition group fixing all of them fixes every generic root and hence is trivial. Thus the residue extension is precisely the splitting field of $Q(t, x)$. $\square$

The congruence class is designed from one certified base specialization: $P(2, x)$ is irreducible modulo 29 and is squarefree with irreducible-factor degrees $(2, 7, 14)$ modulo 11. Taking the modulus $319 = 11 \cdot 29$ freezes both reductions throughout the progression.

For an integer $t \equiv 2 \pmod{319}$, let c_t be the positive greatest common divisor of the coefficients of $P(t, x)$ and put

$$f_t(x) = \frac{1}{c_t} P(t, x) \in \mathbf{Z}[x].$$

Thus f_t is the primitive part of the specialization, with its sign inherited from P . We now prove the second main theorem.

Proof of theorem 1.2. Write $t = 2 + 319n$. Since $319 = 11 \cdot 29$ and every coefficient of P is an integral polynomial in T , one has the polynomial congruences

$$P(2 + 319n, x) \equiv P(2, x) \pmod{11}, \quad P(2 + 319n, x) \equiv P(2, x) \pmod{29}.$$

The leading coefficient of $P(2, x)$ is 5 modulo 11 and 22 modulo 29. Consequently $P(t, x)$ has degree 23, neither prime divides c_t , and passage to the primitive part changes either reduction only by a nonzero scalar.

For a prime not dividing the leading coefficient, write $\bar{f}_{t,p}$ for the monic scalar multiple of the reduction of f_t modulo p . Modulo 29, its coefficient vector, in increasing order and independently of n , is

$$(5, 3, 7, 15, 21, 21, 17, 9, 1, 28, 21, 25, 3, 14, 20, 28, \\ 5, 27, 24, 16, 9, 20, 18, 1).$$

Exact modular powering gives

$$x^{29^{23}} \equiv x \pmod{\bar{f}_{t,29}}, \quad \gcd(\bar{f}_{t,29}, x^{29} - x) = 1.$$

Since 23 is prime, Rabin's criterion proves that $\bar{f}_{t,29}$ is irreducible. In particular, f_t is irreducible over $\mathbf{Q}$; the Dedekind–Frobenius theorem gives a 23-cycle in its Galois group.

Modulo 11, the monic reduction has the factorization

$$\bar{f}_{t,11} = g_2 g_7 g_{14},$$

where

$$g_2 = x^2 + 8x + 3, \\ g_7 = x^7 + 6x^6 + 7x^5 + 6x^4 + 8x^2 + 3x + 1, \\ g_{14} = x^{14} + 8x^{13} + 6x^{12} + 8x^{11} + 6x^{10} + 6x^9 + 3x^8 + 2x^7 \\ + 10x^6 + 3x^5 + x^4 + 8x^3 + 10x^2 + 7x + 6.$$

The three displayed factors are irreducible over $\mathbf{F}_{11}$ and distinct. Thus this reduction is squarefree. Dedekind–Frobenius gives an element with cycle lengths $(2, 7, 14)$, and hence of order 14, in the Galois group.

Because the leading coefficient is nonzero modulo 11, squarefreeness also shows that the discriminant remains nonzero modulo 11; in particular the characteristic-zero discriminant of $P(t, x)$ is nonzero. Applying lemma 6.1 to the regular M_{23} -extension of theorem 1.1 therefore gives

$$H_t := \text{Gal}(f_t/\mathbf{Q}) \hookrightarrow M_{23}.$$

If H_t were proper, it would lie in a maximal subgroup of M_{23} . The ATLAS list [3] gives the maximal-subgroup orders

$$443520, 40320, 40320, 20160, 7920, 5760, 253.$$

Only $253 = 23 \cdot 11$ is divisible by 23, and it is odd. But H_t contains a 23-cycle and an element of order 14, so it cannot be proper. Hence $H_t = M_{23}$.

The finite-field calculations, universal congruences, independent Rabin tests, and maximal-subgroup order check are reproduced in both pure Python and PARI/GP by

$$\text{verification/verify_progression_319.py}, \\ \text{verification/verify_progression_319.gp}.$$

□

Corollary 6.2. *The primitive polynomial $f_2(x) = P(2, x)$ has Galois group M_{23} over $\mathbf{Q}$.*

Its 24 coefficients, ordered from the constant coefficient through the coefficient of x^{23} , are in `data/specialization_t2.Z.json`; the same polynomial is supplied in PARI/GP format. The progression theorem and its witness primes are recorded in `data/progression_319.json`.

6.1. Controlled ramification inside the progression. The generic M_{23} -cover has branch locus

$$\{0, \sqrt{-23}, -\sqrt{-23}\},$$

with local monodromy classes $(2A, 23A, 23B)$, respectively [1, Section 3]. Thus the rational branch point $T = 0$ gives a way to introduce a new prime of ramification without changing the two factorization patterns modulo 11 and 29.

Call a family mutually independent if every finite subfamily is linearly disjoint over $\mathbf{Q}$; equivalently, an r -fold compositum has Galois group M_{23}^r .

Theorem 6.3. *Among the splitting fields of f_t with $t \equiv 2 \pmod{319}$ there is an infinite mutually independent collection of M_{23} -extensions of $\mathbf{Q}$. More precisely, outside a finite set of primes, if $q \nmid 319$ and*

$$t \equiv 2 \pmod{319}, \quad v_q(t) = 1,$$

then the inertia group at q is generated by an element of class $2A$.

Proof. Beckmann's specialization inertia theorem [4, Proposition 4.2] applied at the branch point $T = 0$ says that, away from finitely many bad primes, meeting 0 with intersection multiplicity m produces an inertia group conjugate to $\langle g^m \rangle$, where g is a local monodromy element. Enlarge the bad set to contain 23. If $v_q(t) = 1$, then t meets 0 with multiplicity $m = 1$ and does not meet either root of $T^2 + 23$ modulo q . Thus the inertia group is conjugate to $\langle g \rangle$, with g in class $2A$.

For any such q , the Chinese remainder theorem gives integers satisfying

$$t \equiv 2 \pmod{319}, \quad t \equiv q \pmod{q^2}.$$

The first congruence gives Galois group M_{23} by Theorem 1.2; the second gives $v_q(t) = 1$, hence ramification at q . Inductively choose each new q outside the finite bad set and outside the ramified primes of the fields already constructed. The new field is ramified at q , whereas every preceding field is unramified there, so the fields are pairwise distinct.

It remains to justify mutual independence, since pairwise distinctness alone does not suffice for arbitrary Galois groups. Suppose inductively that $L_1, \dots, L_r$ are linearly disjoint and set $K_r = L_1 \cdots L_r$, so that $\text{Gal}(K_r/\mathbf{Q}) \simeq M_{23}^r$. For a new field L , the intersection $K_r \cap L$ is Galois over $\mathbf{Q}$. Simplicity of M_{23} makes this intersection either $\mathbf{Q}$ or L . In the latter case, restriction would give a surjection $M_{23}^r \twoheadrightarrow M_{23}$. Every such surjection factors through one coordinate: the coordinate images are commuting normal subgroups, and simplicity and the trivial center permit exactly one nontrivial image. The corresponding subfield is then one of the L_i , contradicting the already proved pairwise distinctness. Hence $K_r \cap L = \mathbf{Q}$, completing the induction. $\square$

The construction also admits a completely explicit second field. The following direct discriminant calculation does not require 5 to lie outside Beckmann's exceptional set.

Exact computation 6.4 (A second certified field). Let $E_t = \mathbf{Q}[x]/(f_t)$. The integer

$$3830 \equiv 2 \pmod{319}, \quad 3830 \equiv 5 \pmod{25}$$

lies in the certified progression and meets $T = 0$ 5-adically with multiplicity one. There is also a direct transverse-fiber check over $\mathbf{F}_5$: the repeated part

$$\gcd(\Phi(0, V), \Phi_V(0, V))$$

has degree 8, is squarefree, and is coprime to $\Phi_T(0, V)$. Exact maximal-order computations give

$$v_5(\text{Disc}(E_2)) = 0, \quad v_5(\text{Disc}(E_{3830})) = 8.$$

Consequently the $t = 3830$ splitting field is ramified at 5, while the $t = 2$ splitting field is not, so they are distinct. Here the latter implication uses faithfulness of the natural degree-23 action. Indeed, unramifiedness of E_2 forces inertia in its Galois closure to fix every coset of a point stabilizer; the core of that stabilizer is trivial, so the inertia group is trivial. Conversely, ramification in E_{3830} forces ramification in its Galois closure.

As a consistency check, direct factorization of the discriminant of the minimal-degree plane equation gives, up to a nonzero rational constant,

$$\text{disc}_W P(T, W) = T^8(T^2 + 23)^{22}H_{62}(T)^2, \quad \deg H_{62} = 62.$$

The exponents 8 and 22 agree with the ramification indices of the 2A and 23 fibers; the residual square factor comes from collisions in the singular plane model and is not part of the branch locus of the normalization.

The PARI/GP certificate reproduces the transverse-fiber check, discriminant factorization, and maximal-order calculation:

`verification/verify_ramified_t3830.gp.`

The dependency-free companion independently checks the CRT conditions and both finite-field factorization patterns:

`verification/verify_ramified_t3830.py.`

The machine-readable record is `data/ramified_t3830.json`.

7. SCOPE, PROVENANCE, AND DISCLOSURE

The M_{23} theorem, the original degree-(23, 8) equation, and the minimality argument are due to Huang et al. [1]; the contribution here is the explicit minimal-degree function, its equation, the adjoint recovery, exact certificates of equality with their function field, and the certified arithmetic progression and controlled-ramification refinement in theorems 1.2 and 6.3. Arithmetic progressions in Hilbert sets and specializations with prescribed local behavior are part of the general Hilbert–Grunwald theory [5]; in particular, full-group linearly disjoint specializations with prescribed inertia in arithmetic progressions are covered by [6]. The claim here is the explicit equation, displayed modulus, and uniform certificate for this cover, not a new general specialization principle.

The discovery and drafting process used substantial AI assistance.

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